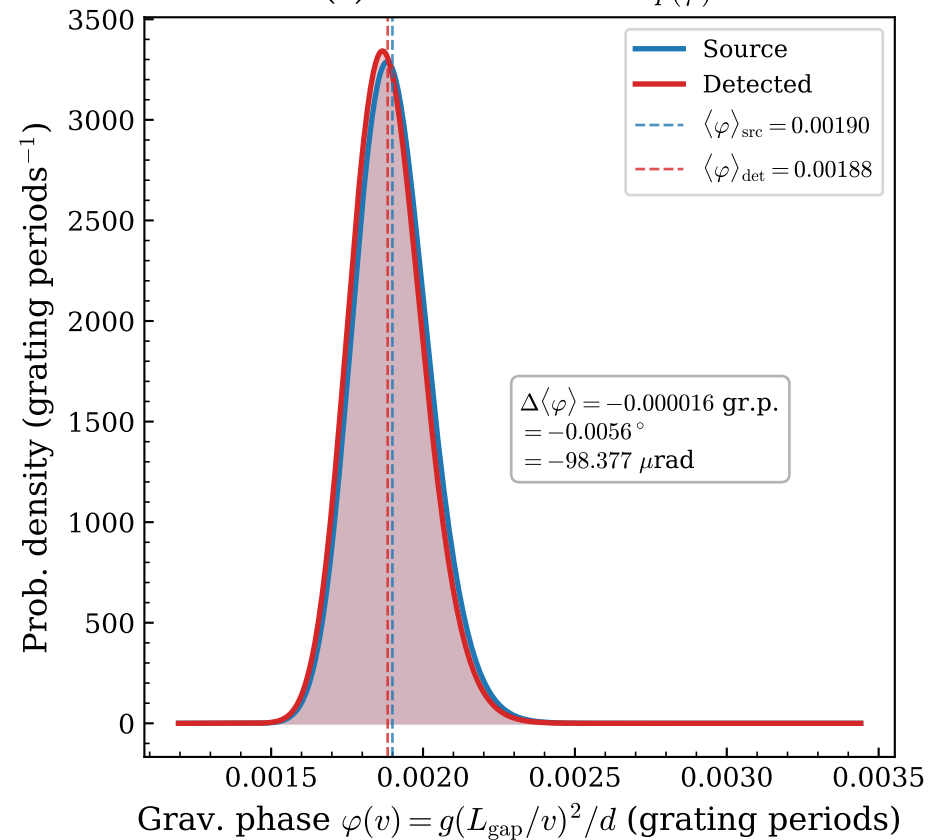


Gravitational phase shift from lifetime selection — $\varphi(v) = g(L_{\text{gap}}/v)^2/d$, $v_0 = 2175$ m/s, $\sigma_v = 70$ m/s, $L_{\text{gap}} = 9.56$ mm, $d = 100$ nm, $\tau_\mu = 2.1970$ μs

(a) Phase distribution $p(\varphi)$



(b) Integrated fringe $I(x) = \int p(v) \cos[2\pi(x/d - \varphi(v))] dv$

